Rank spectra, permutation orbits and almost simple groups

An attributed synthesis of the fixed classification survey

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We consider the twenty-one-page edition of Leemans's survey [1], arXiv:1910.08843v2, 11 May 2020. It contains thirty-one numbered theorems, one corollary, six tables, one diagram figure, fifteen problems and three conjectures. A survey has a different logical contract from a classification paper: most of its theorems are statements of earlier results. We therefore distinguish the proofs acquired in this project from the published classification theorems that remain explicit inputs. This distinction applies to every row of the accompanying clause register. Recovering this synthesis does not confer recovery credit on any other original paper.

Our companion *Three elementary interfaces for a survey of string C-groups* proves the full rank-three intersection theorem, the symmetric rank constructions with their exact corrected types, and two counterexamples to the literal survey Theorem 5.7. Here we assemble those results with the acquired uniform theories, record the exact numerical conventions, and give an explicit disposition for every source claim and question. We use "acquired" only for a complete proof with its recorded independent review and acceptance; it does not mean an unproved statement inferred from a matching numerical answer.

1 Three principles and their actual hypotheses

For an ordered involution marking $\rho = (\rho_0, \dots, \rho_{r-1})$ write $G_I = \langle \rho_i : i \in I \rangle$. Its intersection property is

$$G_I \cap G_J = G_{I \cap J} \qquad (I, J \subseteq \{0, \dots, r-1\}).$$

The string condition requires distant marks to commute. In all uses of a subgroup, the marking is the actual subgroup marking. Passing to an orbit or a block action is a homomorphism and does not, by itself, preserve the intersection property.

The following three acquired principles account for much of the survey. First, a faithful action of a finite group H is the disjoint union of coset actions H/K, with conjugacy classes of stabilizers and their multiplicities as complete invariants. Its kernel is the intersection of the corresponding cores. Stabilizers of a partial marking act on possible next marks, so successive orbit selection gives a complete census when every admissible extension is tested. The acquired finite census proofs establish this induction, the full intersection test, and the bound $r \le \lfloor \log_2 |G| \rfloor$ used for termination.

Second, the acquired restriction-kernel theorem controls adjoining a leaf to a permutation marking. It tests the actual kernels of prefix restrictions, rather than assuming an extension preserves intersections. For high-rank symmetric markings, the pure-cut theorem strengthens this to an invertible subdivision operation, compatible with isomorphism and reversal. It proves stability, not merely the construction of additional examples.

Third, the acquired contraction theorem says that an irreducible string C-marking of rank $r \ge 4$ contracts to

$$(\rho_1, \rho_0\rho_2, \rho_3, \dots, \rho_{r-1})$$

whenever $\rho_0 \in \langle \rho_0\rho_2, \rho_3 \rangle$. If $p_j = |\rho_{j-1}\rho_j|$, oddness of p_3 suffices: the corresponding odd power of $(\rho_0\rho_2)\rho_3$ recovers ρ_0. The complete acquired proof pays generation and every intersection.

Repeatedly testing $p_3, p_4, \ldots, p_{t+2}$ gives all ranks from $r-t$ through r, for $0 \le t \le r-3$. No step tests p_2. The extra initial test in survey Corollary 1 can leave its displayed maximum undefined. The corrected iteration permits zero steps without an oddness assumption and implies every defined instance of the printed conclusion. Reuse

Lemma 1. *A C-marking of an almost simple group has connected Coxeter diagram.*

Proof. Let S be its nonabelian simple socle. A disconnected diagram and the intersection property would give a direct product $G = A \times B$ of nontrivial normal subgroups. Since $A \cap S$ is normal in S, it is either S or 1. In the first case B centralizes S; in the second $[A, S] \le A \cap S = 1$, so A centralizes S. But $S \le G \le \mathrm{Aut}(S)$ has $C_G(S) = 1$. Both alternatives are impossible. □

For rank three, the companion proves the theorem of Conder and Oliveros: a finite nontrivial group with no nontrivial cyclic normal subgroup, generated by involutions a, b, c with $ac = ca$, has the full intersection property. The argument includes the exceptional Klein-four intersection of its two dihedral subgroups. Thus the rank-three positive simple-group classification reduces to involution generation, subject to its genuinely substantial published input. We retain the Nuzhin–Mazurov classification, with the Macaj–Jones corrections, under its original attribution. The acquired simple-group obstruction separately proves that the twelve exceptional families listed in survey Theorem 3.1 have no string C-marking of any rank.

2 Symmetric and alternating groups

The adjacent transpositions along a path generate S_n. A selected set generates the direct product of the symmetric groups on the path components; intersections correspond to common selected edges. Thus this is a C-marking. Contraction gives every rank $3 \le r \le n-1$. The companion computes the exact types:

$$\begin{cases} [3^{n-2}], & r = n-1, \\ [n-r+2, 6, 3^{r-3}], & 3 \le r \le n-2. \end{cases} \tag{1}$$

The second line cannot include the first case, as the literal survey Theorem 4.3 does: already S_4 has no element of order six. This corrects the type, not the rank-existence theorem. Correction

The acquired symmetric stability theorem proves that the number of string markings of rank $n-k$, modulo isomorphism and duality, is independent of n for $n \ge 2k+3$. Its first six values are

$$1,\ 1,\ 7,\ 9,\ 35,\ 48. \tag{2}$$

The proof is the complete pure-cut bijection, together with the complete rooted-wing calculation at the boundary. It resolves survey Conjecture 1 as a consequence of the later original [2]; this is a recovered previous result, not a new solution to an open conjecture. The separate acquired diagram classification for $1 \le k \le 4$ matches all eighteen diagrams of the survey's Figure 1. The original TeX arrows, parallel edges and ellipses were parsed with the unchanged accepted source adapter. Exact point conjugacies to all eighteen reviewed family models are retained. The uniform subdivision proof then covers every parameter, including the smaller boundary where the printed long ellipses do not literally fit.

For A_n, the acquired complete theorem on maximal string rank gives

$$\max r = \begin{cases} 3, & n = 5, \\ 4, & n = 9, \\ 5, & n = 10, \\ 6, & n = 11, \\ \lfloor (n-1)/2 \rfloor, & n \ge 12. \end{cases} \tag{3}$$

There is no string C-marking for $n = 3, 4, 6, 7, 8$. The acquired construction and contraction proof gives every rank $3 \le r \le \lfloor (n-1)/2 \rfloor$ when $n \ge 12$. In particular it implies the existence conclusion "some alternating group for every rank" in survey Theorem 4.6. Its two more precise displayed families,

$$A_{2r+1} : [10, 3^{r-2}] \ (r \ge 4 \text{ even}), \qquad A_{2r+3} : [10, 3^{r-4}, 6, 4] \ (r \ge 5 \text{ odd}), \tag{4}$$

remain explicitly the constructions of Fernandes–Leemans–Mixer [3]. Their exact types and diagrams are not inferred from an existence proof for another marking. The first is already an input of the acquired alternating interval proof; its other maximal seed is different from the second family in (4). This distinction prevents an accidental transfer of an unproved Schlaefli type.

Pairs of commuting involutions

Reuse

Two distinct nonidentity commuting involutions generate $V = C_2^2$. Let f, x, y, z, t be the multiplicities of its orbits with stabilizers $V, \langle a \rangle, \langle c \rangle, \langle ac \rangle, 1$, respectively. A degree-N pair is faithful precisely when

$$N = f + 2(x + y + z) + 4t, \qquad t > 0 \text{ or at least two of } x, y, z > 0. \tag{5}$$

Exchanging the involutions exchanges x, y. Both involutions are even precisely when $x \equiv y \equiv z \pmod 2$. Thus counting (5) with $x \le y$ proves the two formulas of survey Theorems 4.4 and 4.10, by the complete acquired multiplicity calculation. At degree six, full automorphism fusion replaces the permutation-conjugacy answers by five classes for S_6 and one for A_6. We retain that complete finite automorphism calculation, not an exception inferred from the source table. All fourteen rows of survey Table 2 agree with these formulas; a separate literal orbit count agrees for every degree at most fifty. The uniform proof is the orbit classification (5).

3 Linear and classical groups

The complete acquired subgroup-separator proofs recover the following string classifications: $\mathrm{PSL}_2(q)$ has rank at most four, with rank four precisely at $q = 11, 19$, yielding respectively the unique 11-cell and 57-cell; $\mathrm{PGL}_2(q)$ has rank at most four, with rank four precisely at $q = 5$, yielding the simplex. The standard geometry and presentation identifications remain named inputs of those proofs. The rank-three existence classification of Sjerve–Cherkassoff is retained: $\mathrm{PSL}_2(q)$ occurs exactly for $q \notin \{2, 3, 7, 9\}$, and $\mathrm{PGL}_2(q)$ occurs exactly for $q \ne 2$.

The larger classification in survey Theorem 5.6 is a separate input: Connor–De Saedeleer–Leemans [5] give the group classification and rank assertions for almost simple groups with PSL_2 socle. Its small cases, even-field cases, odd fields of odd exponent, and Baer extensions are retained exactly in the clause register. Neither the string results for $\mathrm{PSL}_2, \mathrm{PGL}_2$ alone nor the project's finite semilinear chiral census proves that infinite theorem. We retain its published proof cost rather than hiding it in a label such as "standard classification". Correction

The unnamed hypermap assertion at the start of Section 5.4 is false. The separately checked companion gives explicit matrices in $\mathrm{PSL}_2(11)$ defining a regular hypermap of type $(6, 6, 5)$. Its three distinguished involutions generate the full group, while $|\langle a, b \rangle \cap \langle b, c \rangle| = 4 > 2 = |\langle b \rangle|$. This example lies in the hyperbolic, inner-reflexible range actually classified by Conder–Potočnik–Širáň [9]. The missing condition in the survey's appeal to its Theorem 3.2 is the commutation of two distinguished involutions. The valid string theorem and the original hypermap classification are unaffected. This is a correction of an unnamed inference, not a newly settled open question. Correction

Survey Theorem 5.7, stated for an arbitrary subgroup of $\mathrm{GL}_3(q)$, is false. The companion's faithful S_4 action on $\{x \in \mathbb{F}_q^4 : \sum x_i = 0\}$ has the simplex C-marking in characteristic two, contradicting the unqualified claim that q is odd. Over $\mathbb{F}_5$ the same action is absolutely irreducible,

but its order 24 is smaller than $|\Omega_3(5)| = 60$. It therefore also refutes the displayed lower containment, even after adding irreducibility. The body Theorem 3.3 of the cited Brooksbank–Vicinsky manuscript states form preservation, without that lower containment.

The acquired corrected projective theorem suffices for the intended application. An irreducible projective string C-group in dimension three has rank at most four, is in odd characteristic, and preserves a nondegenerate form after lifting. The natural projective actions of $\mathrm{PSL}_3(q)$ and $\mathrm{PGL}_3(q)$ are irreducible. In odd characteristic

$$\frac{|O_3(q)|}{|\mathrm{PSL}_3(q)|} = \frac{2\gcd(3, q-1)}{q^2(q^3-1)} < 1,$$

so neither full projective group can occur. This is an actual form obstruction, not a lower-containment assertion about every subgroup. For $\mathrm{PSL}_4(q)$, survey Theorem 5.8 retains the published odd-field rank-four construction of Brooksbank–Leemans [6]; the acquired simple-group obstruction excludes every even field. The previously recorded project work on the maximum rank is kept separate in Section 6.

For $q = 2^k \geq 4$, the acquired quadratic-path construction gives both $O^+(2m, q)$ and $O^-(2m, q)$ in rank $2m$, and $\mathrm{Sp}(2m, q)$ in rank $2m + 1$, including $m = 1$. Its generation and intersection proofs use supported symmetry groups on actual subspaces. Free nonzero endpoint coefficients choose both orthogonal types and a nonsingular odd-dimensional radical. In the notation of that proof, an adjacent product has characteristic polynomial

$$X^2 + \alpha^2 X + 1, \qquad \alpha \neq 0.$$

It has distinct reciprocal roots in $\mathbb{F}_q$ or $\mathbb{F}_{q^2}$, and its order divides $q - 1$ or $q + 1$. Thus every adjacent order is odd, including both free endpoints. Contraction consequently gives all the rank intervals in survey Theorem 10.3. The earlier stronger assertion that every chosen bond has order exactly $q+1$ is unnecessary and is not retained for the corrected endpoint construction.

4 Suzuki groups, Ree groups and arbitrary diagrams

The acquired Suzuki argument gives a particularly short example of how subgroup structure and orbit counting complement one another. The outer quotient of $\mathrm{Aut}(\mathrm{Sz}(2^N))$, for odd N, has odd order. An involution-generated intermediate group therefore lies in the simple socle. The acquired involution-centralizer and dihedral arguments give precisely string rank three. Normalizing two commuting endpoint involutions reduces the possible third marks to a field parameter action. Exact subfield support, followed by field Moebius inversion, removes the proper subfield groups. Orbit counting and duality give precisely the formula in survey Theorem 6.2. All Suzuki subgroup and generation theorems used in that derivation remain explicit acquired inputs.

For small Ree groups, the complete acquired proof gives string rank three for $R(3^N)$ with odd $N > 1$, excludes larger ranks and excludes the nonsimple $R(3)$. The odd outer quotient again excludes proper almost simple extensions. The recovery replaces the source's false normality step and keeps the exact subgroup and fusion hypotheses; it does not transport the Suzuki proof by analogy.

Removing the string condition changes the problem. We retain as published inputs the full arbitrary-C-group classification of Suzuki groups by Connor–Leemans [7], including existence of a nonstring marking, and the full PSL_2 / PGL_2 classification by Connor–Jambor–Leemans [8]. The latter states that the *maximum* C-rank of $\mathrm{PSL}_2(q)$, $q \geq 4$, is four exactly for

$$q \in \{7, 9, 11, 19, 31\},$$

and otherwise three; for $\mathrm{PGL}_2(q)$ it is four exactly at $q = 5$. Survey Theorem 11.2 instead says that a representation has rank four if and only if its parameter is in this list. That literal Correction

universal reading is false: $\mathrm{PSL}_2(11)$ and $\mathrm{PGL}_2(5)$ also have rank-three string markings, hence rank-three C-markings. The cited authors' publisher and conference abstracts explicitly state the maximal-rank interpretation used here.

The two final symmetric C-group classifications are acquired in full. For degree $n \geq 7$ and rank $n-1$, the marks are precisely the edges of a tree, so they are transpositions and generate S_n; the Coxeter diagram is the line graph of that tree. For rank $n-2$ in S_n, $n \geq 9$, the three families of survey Theorem 11.4 are exactly those of the acquired complete high-rank extension. Its necessity and sufficiency arguments allow arbitrary branching in the attached tree. The earlier incomplete version of that application is not used.

5 The half-degree theorem and numerical conventions

The acquired transitive-group theorem recovers all alternatives of survey Theorem 10.1, including its complete degree-six and degree-eight imprimitive exceptions and the primitive degree-six simplex. In the infinite equality family, put $n = 2m$ with m odd. Write λ for the simultaneous flip of all m pairs, e_1 for one pair flip, and s_i for the adjacent transpositions of pairs. The actual tuple

$$(\lambda, \lambda + e_1, s_1, \ldots, s_{m-1})$$

has type $[2, 4, 3, \ldots, 3]$. Reversing only the noncentral path gives $[2, 3, \ldots, 3, 4]$. These are generally distinct ordered polytopes up to global duality, although their unordered distinguished set is the same. The survey prints the latter type beside the former tuple. Its Table 6 already has the correct $[3, 2, 3]$ type for $S_3 \times S_3$; we do not attribute the different typo in the earlier paper to this survey edition.

For the simple groups and connected symmetric markings in Tables 1, 3 and 5, the noncommutation graph determines the path ordering up to reversal. If a complete census has c_r ordered classes and s_r classes fixed by reversal, the survey convention is

$$d_r = (c_r + s_r)/2. \tag{6}$$

This follows by partitioning the classes into fixed points and unordered two-element reversal orbits. It is not valid to halve c_r without also counting the self-dual classes.

The numerical appendix transcribes every numeric cell directly from the fixed source TeX and records a disposition for each cell. The completed L071 censuses recover every survey row through the Held group, the entire Co_3 row, the S_7, S_9, A_5, A_9 rows, and the $A_{11}, \ldots, A_{14}$ rows after applying (6). The exact recorded self-dual counts are used, with no repeated search. The symmetric stability theorem supplies every cell in its range, and the complete maximal-rank theorems supply the zero cells above their bounds. All fourteen rows of Table 2 follow from (5) and the exceptional automorphism action. These are statements about named cells, not a blanket validation of a table.

The complete retained A_{10} lists have $177, 4, 8$ ordered classes in ranks $3, 4, 5$. Direct reversal matching gives $11, 0, 0$ fixed classes, hence the printed values $94, 2, 4$. The same calculation on the complete S_5 list gives seven ordered rank-three classes and one fixed class, hence four after reversal. The degree-six automorphism calculation gives the two rank-three S_6 classes. The separately acquired complete six-point component now also verifies its middle value, four in rank four, using the full outer-automorphism action. The complete S_8 component gives ordered counts $126, 71, 20, 2, 1$ and reversal fixed counts $10, 1, 2, 0, 1$. Thus its source row is $68, 36, 11, 1, 1$; all four initial classes and all 71 296 actual subset intersections are independently checked. The retained A_{10} and S_5 calculations reuse their completed lists; the new S_6 and S_8 components execute the acquired common method on precisely their fixed rows. The complete S_{10} component now gives ordered counts $809, 403, 99, 26, 13, 2, 1$ and reversal fixed counts $17, 3, 5, 0, 1, 0, 1$. Its source row is therefore $413, 203, 52, 13, 7, 1, 1$. All five initial involution classes finish, and a separate actual-group verification checks every one of the 969 024 subset intersections. The four entries in ranks

three through six are newly acquired here; the three higher entries were already consequences of stability. Likewise, the complete S_{11} calculation has ordered counts $2382, 377, 86, 49, 18, 13, 2, 1$ and reversal fixed counts $60, 1, 0, 1, 0, 1, 0, 1$. Its row is $1221, 189, 43, 25, 9, 7, 1, 1$. The complete five-class search and all 3 257 472 actual subset intersections have passed distinct review. Only ranks three through six add new cells; the four higher entries were already supplied by stability.

The last four rows are supplied by the companion *Four finite rows of the classification survey.* Its complete search has twenty-two initial involution classes. Every branch terminates; its 94 354 actual tuples are checked by 150 173 subgroup intersection tests. These tests use the acquired interval criterion: a string marking has the full intersection property if its two end subgroups do, and their intersection is exactly the common middle subgroup. Induction over contiguous intervals, starting with ranks one and two, proves the full property. Thus the reduction in verification work is a theorem, not the omission of selected subset tests. An independent literal conjugacy calculation pays the reversal convention. The finite results are as follows; entries in each row start at rank three.

Group	Ordered classes	Classes modulo duality
S_{12}	$6567, 1875, 364, 150, 74$	$3346, 940, 183, 75, 40$
S_{13}	$14181, 1723, 341, 244, 82$	$7163, 863, 171, 123, 41$
S_{14}	$45891, 7869, 1951, 606, 324, 108$	$23126, 3945, 978, 303, 163, 54$
A_{15}	$11557, 314, 40, 81, 12$	$5820, 158, 20, 42, 6$

The reversal fixed counts are respectively $(125, 5, 2, 0, 6)$, $(145, 3, 1, 2, 0)$, $(361, 21, 5, 0, 2, 0)$ and $(83, 2, 0, 3, 0)$. Higher symmetric ranks follow from (2), in its actual stable range; the alternating maximum (3) excludes all ranks above seven. Every one of the twenty-one previously unverified experimental cells is therefore established without taking a target number as a premise.

There are 221 displayed rank-count cells across Tables 1, 3 and 5. Of these, 212 now have an acquired project proof or complete census; the remaining nine use the separately attributed published classification in the next paragraph. This total does not subsume the different data in the other tables. Table 2 has fourteen formula rows, Table 4 has the two infinite constructions (4), and Table 6 has four finite group/type rows. For the degree-six group of order 48, Table 6 does not exhaust the possible ordered types: the acquired list also gives $[2, 4, 3]$ and its reversal. Each printed row is matched to a literal accepted tuple.

The Rudvalis, sporadic Suzuki and O'Nan entries refer to named classification papers. The published Leemans–Mulpas paper [10], Section 5, states the complete counts $21594, 227$ for Rudvalis, $7119, 257, 13$ for Suzuki, and $6536, 16$ for O'Nan, in successive ranks starting at three, with no higher ranks. These are exact attributed classification inputs; our separate recovery of that manuscript is not yet complete. In particular, the literal "Unknown" in the survey's O'Nan rank-three cell was answered in that 2022 publication. It is preserved as historical source text, not presented as a presently open question or as a count computed in this companion. The regular-map count is a different quantity: the project's bounded correction to it does not prove the O'Nan string count. Finally $|M_{22}| = 443520$; Table 5's 443510 is a literal order typo, verified against the acquired exact group model.

6 Questions, attribution and the recovery boundary

Every one of the eighteen numbered problems or conjectures has an individual record. Conjecture 1 is answered by the acquired later stability theorem, with its original authorship retained. General symmetric enumeration (Problem 1) is therefore partly answered, not entirely solved. The construction and maximum-rank theorems for alternating groups do not enumerate them (Problem 2).

The project already has internally reviewed written work on the odd-q maximum-four conjecture for $\mathrm{PSL}_4(q)$ (Conjecture 2), and on part of Problem 7. Those exact earlier arguments,

checks, reviews and priority qualifications are linked as historical project results. They are not assigned another conjecture credit here. Problem 7 includes field and graph extensions, whereas the recorded diagonal interval results do not cover those extensions. The unitary companion results are likewise not silently substituted for a linear-socle question. No claim of historical priority follows from the earlier bounded literature searches.

Problems 3–6 concern complete extension spectra or enumeration and are not settled by the linear-socle obstruction. Problem 8 is a Ree enumeration problem. Problem 9 asks for maximal ranks of the characteristic-two classical groups; construction of the displayed rank intervals alone does not prove maximality. Problem 10, the O'Nan rank-three computation, has the published answer 6536 in the 2022 classification just cited; it is not a new result of this project. Problem 11 asks for the wider remaining sporadic classification and is not settled by those three named groups alone. Problems 12–15 concern arbitrary C-groups, whose questions are not answered by string classifications. Conjecture 3, that A_{11} is the only finite simple group with a gap in its rank spectrum, is not proved by certifying that one gap. These are dispositions of the fixed source questions, not a new search asserting their present-day openness.

The new arguments in this survey integration include the companion's complete cyclic-normal intersection proof, the explicit counterexamples to the dimension-three and hypermap assertions, and the complete finite classifications just stated. Contraction, orbit multiplicities, pure-cut stability, supported symmetry groups and the completed censuses are reused with their previous authorship and reviews. The published external classification premises are listed separately. Neither citations nor finite agreement are treated as reproofs. Of the thirty-two numbered theorem and corollary clauses, twenty-five have an acquired proof, correction or counterexample route, with their stated prior inputs. The other seven—Theorems 3.1, 4.6, 5.1, 5.4, 5.6, 11.1 and 11.2—retain a published classification or exact construction. Their statements and attribution are part of this synthesis; their proof cost is not silently transferred to a newly proved theorem. The nine sporadic numerical cells have the same explicit status. No additional Lean claim or numerical compression factor is made at this intake stage.

Complete theorem index

Source	Acquired route or retained input	Exact qualification
3.1	Simple-group obstruction; Nuzhin–Mazurov and corrections	Positive classification retained; all-rank negative obstruction acquired.
3.2	Companion intersection theorem	Full no-cyclic-normal-subgroup hypothesis, including the Klein-four case.
4.1,4.3	Contraction and path construction	All ranks; endpoint type corrected in (1).
4.2	Pure cuts and 18 family models	Figure 1, stable range and connected diagram; exact source arrows matched.
4.4,4.10	Orbit multiplicities	Exact formulas, parity and full degree-six automorphism fusion.
4.5	Positive classification; alternating constructions	Rank-three existence, with retained published classification where used.
4.6	Alternating interval; 2012 seeds	Existential conclusion acquired; the two exact types in (4) retained explicitly.
4.7–4.9	Alternating interval and maximum theorem	Exact small groups and every rank in the stated interval; nontrivial degree domain.
5.1,5.4	Sjerve–Cherkassoff	Complete rank-three classification retained; no substitution of a finite sample.

Source	Acquired route or retained input	Exact qualification
5.2,5.3,5.5	Projective subgroup separators	Full rank-four classification with standard cell identifications retained.
5.6	Connor–De Saedeleer–Leemans	Complete almost-simple classification is a named external input.
5.7	Companion and dimension-three obstruction	Literal statement refuted; correct irreducible form obstruction retained.
5.8	Simple-group obstruction; Brooksbank–Leemans	Even fields excluded; published odd-field rank-four construction retained.
6.1,6.2	Suzuki normalization and field support	Exact rank theorem and both levels of Moebius inversion.
7.1	Complete small-Ree proof	Actual subgroup/fusion hypotheses and repaired normality argument.
8.1	Supported symmetries on quadratic paths	Both types and odd-dimensional quotient; corrected free endpoints.
10.1	Complete transitive half-degree theorem	Literal wreath tuple/type distinguished from reordering and global duality.
10.2	Actual subgroup contraction	Membership hypothesis retained.
Cor.1	Exact contraction iteration	Tested bonds start with p_3; zero-step case always permitted.
10.3	Quadratic paths plus contraction	All adjacent orders odd, sufficient for every stated rank.
11.1	Connor–Leemans	Full arbitrary-C-group Suzuki theorem retained, including nonstring existence.
11.2	Connor–Jambor–Leemans	Maximal C-rank classification retained; universal wording corrected.
11.3	Extremal independent markings	Full tree necessity and sufficiency for $n \geq 7$.
11.4	Complete high-rank symmetric extension	All three families with arbitrary attached-tree branching.

References

[1] D. Leemans, *String C-group representations of almost simple groups: a survey*, arXiv:1910.08843v2, 11 May 2020; Contemp. Math. 764 (2021), 157–178. All source numbers in this paper refer to that fixed version.

[2] P. J. Cameron, M. E. Fernandes and D. Leemans, *The number of string C-groups of high rank*, fixed arXiv:2212.12723v2. Acquired complete corrected principal recovery L019.

[3] M. E. Fernandes, D. Leemans and M. Mixer, *Polytopes of high rank for the alternating groups*, J. Combin. Theory Ser. A 119 (2012), 42–56, Theorem 1.1. The exact seeds in (4) remain inputs, including generation and full intersection property.

[4] D. Sjerve and M. Cherkassoff, *On groups generated by three involutions, two of which commute*, CRM Proc. Lecture Notes 6 (1994), 169–185. Retained precise existence classification.

[5] T. Connor, J. De Saedeleer and D. Leemans, *Almost simple groups with socle PSL(2,q) acting on abstract regular polytopes*, J. Algebra 423 (2015), 550–558. Retained full classification.

[6] P. A. Brooksbank and D. Leemans, *Polytopes of large rank for PSL(4,q)*, J. Algebra 452 (2016), 390–400, Theorem 1.1. Retained odd-field rank-four existence.

[7] T. Connor and D. Leemans, *C-groups of Suzuki type*, J. Algebraic Combin. 42 (2015), 849–860, Classification Theorem 1.1.

[8] T. Connor, S. Jambor and D. Leemans, *C-groups of PSL(2,q) and PGL(2,q)*, J. Algebra 427 (2015), 455–466, Theorem 1.1. Retained maximal-rank and explicit rank-four classification.

[9] M. Conder, P. Potočnik and J. Širáň, *Regular hypermaps over projective linear groups*, J. Aust. Math. Soc. 85 (2008), 155–175. The inner-reflexible correspondence is retained; the blanket transfer to C-group markings is refuted.

[10] D. Leemans and J. Mulpas, *The string C-group representations of the Suzuki, Rudvalis and O'Nan sporadic groups*, Art Discrete Appl. Math. 5 (2022),#P3.09, Section 5; doi:10.26493/2590-9770.1405.4ce. Exact external census inputs; no independent re-exhaustion is claimed in this survey companion.

[11] D. Leemans and M. Mixer, *Algorithms for classifying regular polytopes with a fixed automorphism group*, Contrib. Discrete Math. 7 (2012), 105–118. The complete acquired L071 recovery includes the earlier Hartley–Hulpke lists and exact duality counts.